

Spatial Descriptions & Transformations

Chapter 2

Spatial Descriptions & Transformations

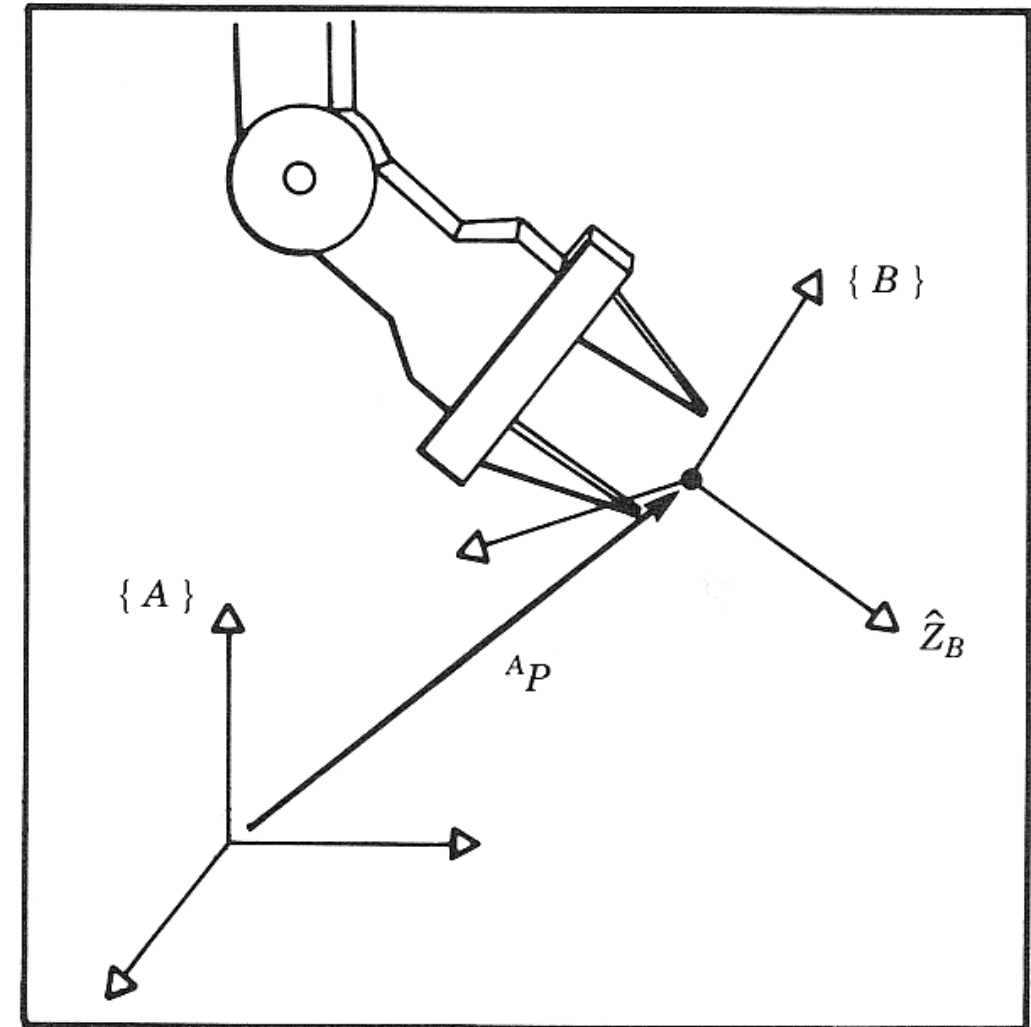
Central Topic

Problem

Robotic manipulation, by definition, implies that parts and tools will be moving around in space by the manipulator mechanism. This naturally leads to the need of *representing positions and orientations of the parts, tools, and the mechanism itself.*

Solution

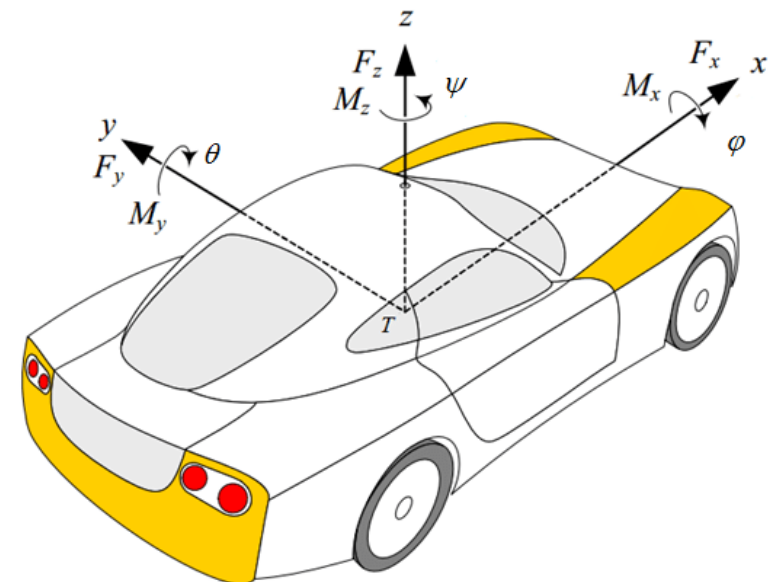
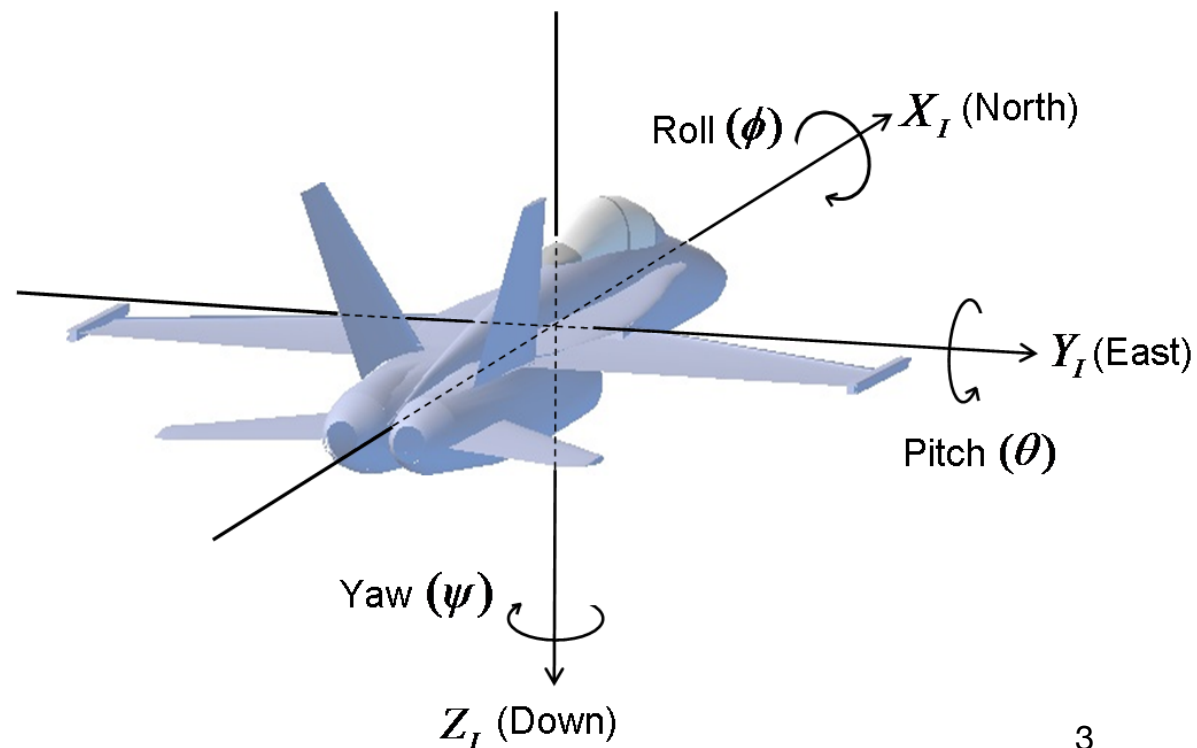
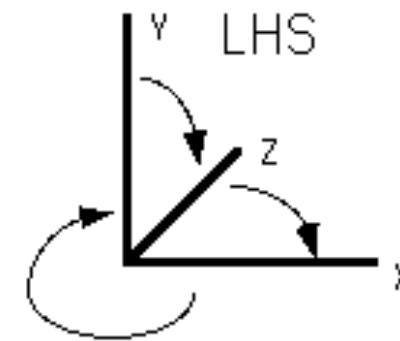
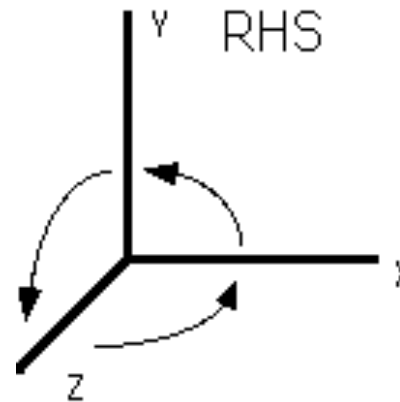
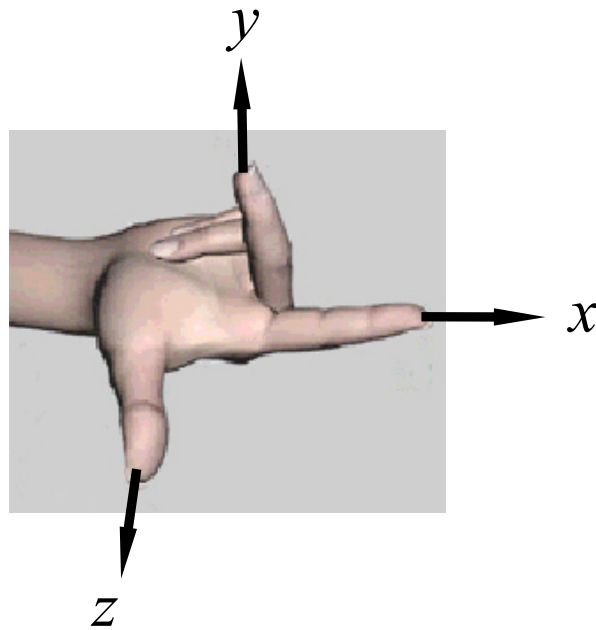
Mathematical tools for representing position and orientation of objects / frames in a 3D space.



Frames, Mapping, Operator, Homogeneous Transformation, Inverse Transformation

Spatial Descriptions & Transformations

Coordinate System



Spatial Descriptions & Transformations

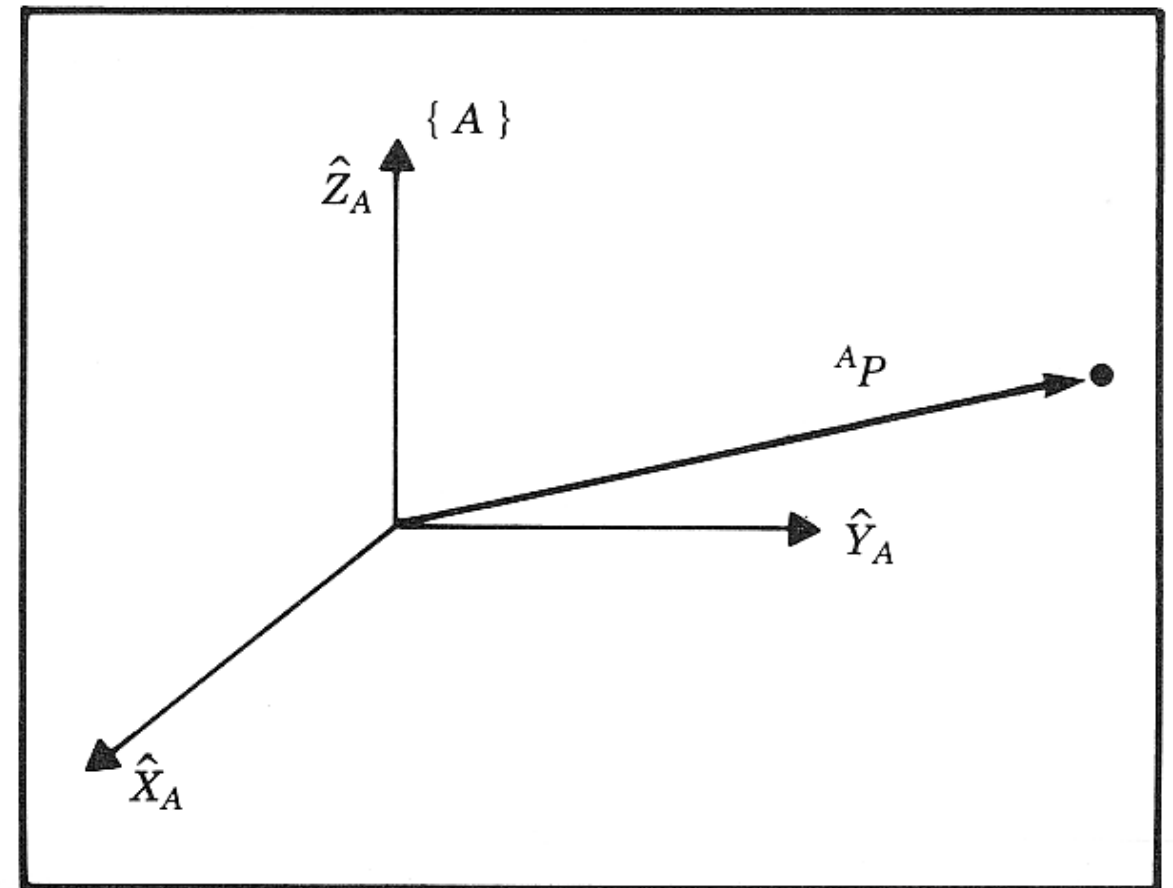
Description of a Position

The location of any point in can be described as a 3x1 **position vector** in a reference coordinate system

Coordinate System

$${}^A P = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

Position Vector



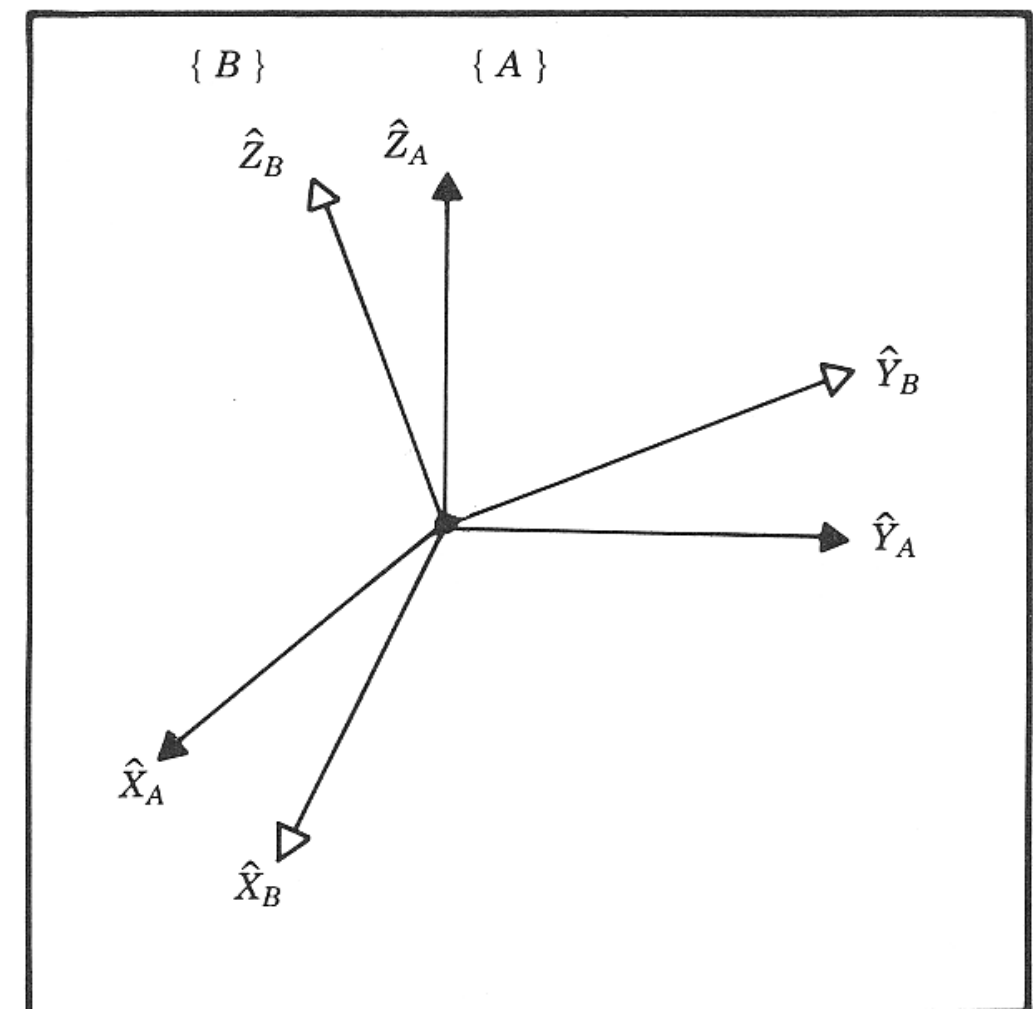
Spatial Descriptions & Transformations

Description of an Orientation

The orientation of a body is described by attaching a coordinate system to the body {B} and then defining the relationship between the body frame and the reference frame {A} using the **rotation matrix**

The rotation matrix describing frame {B} relative to frame {A}

$$\begin{matrix} \text{Reference Frame} \\ {}^A_B R = [{}^A\hat{X}_B, {}^A\hat{Y}_B, {}^A\hat{Z}_B] = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \\ \text{Body Frame} \end{matrix}$$



Spatial Descriptions & Transformations

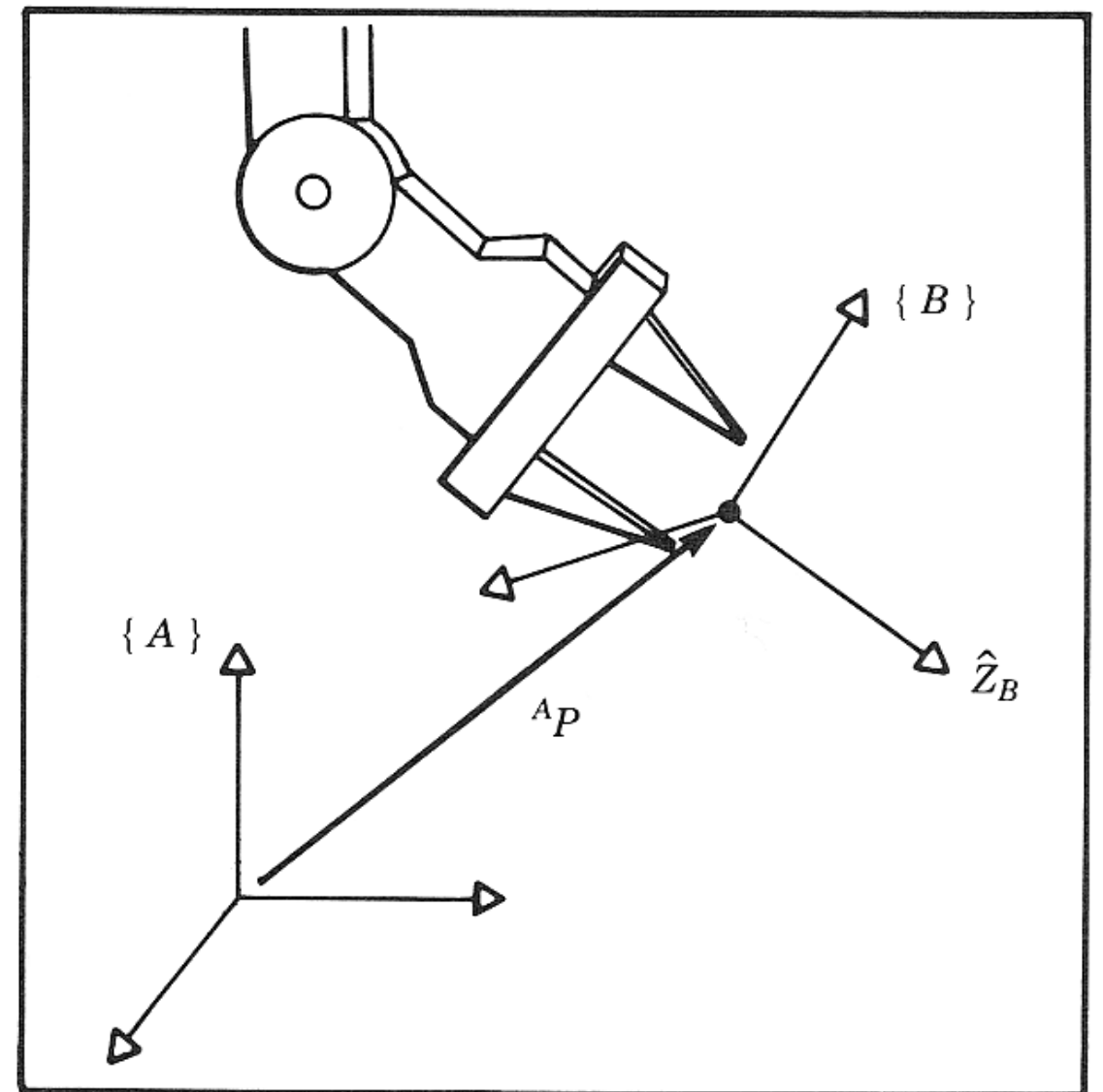
Description of a Frame

The information needed to completely specify where an object is in 3D space, both position and orientation

The rotation matrix describing frame {B} relative to frame {A}

$$\{B\} = \{ {}^A_B R, {}^A P_{BORG} \}$$

The origin of frame {B} relative to frame {A}

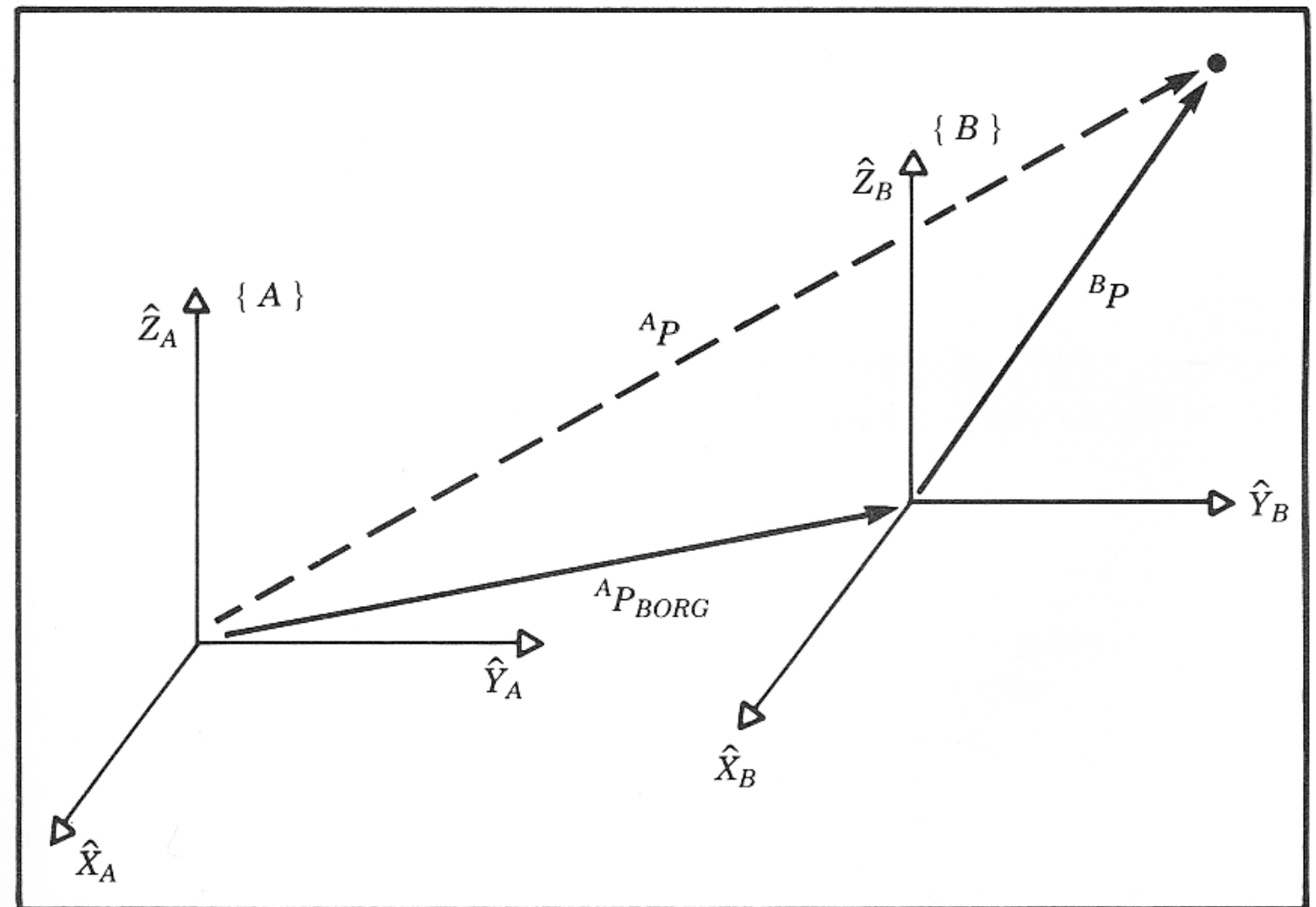


Spatial Descriptions & Transformations

Mapping: Translated Frames

Assuming that frame {B} is *only translated* (not rotated) with respect frame {A}.
The position of the point can be expressed in frame {A} as follows.

$${}^A P = {}^B P + {}^A P_{BORG}$$



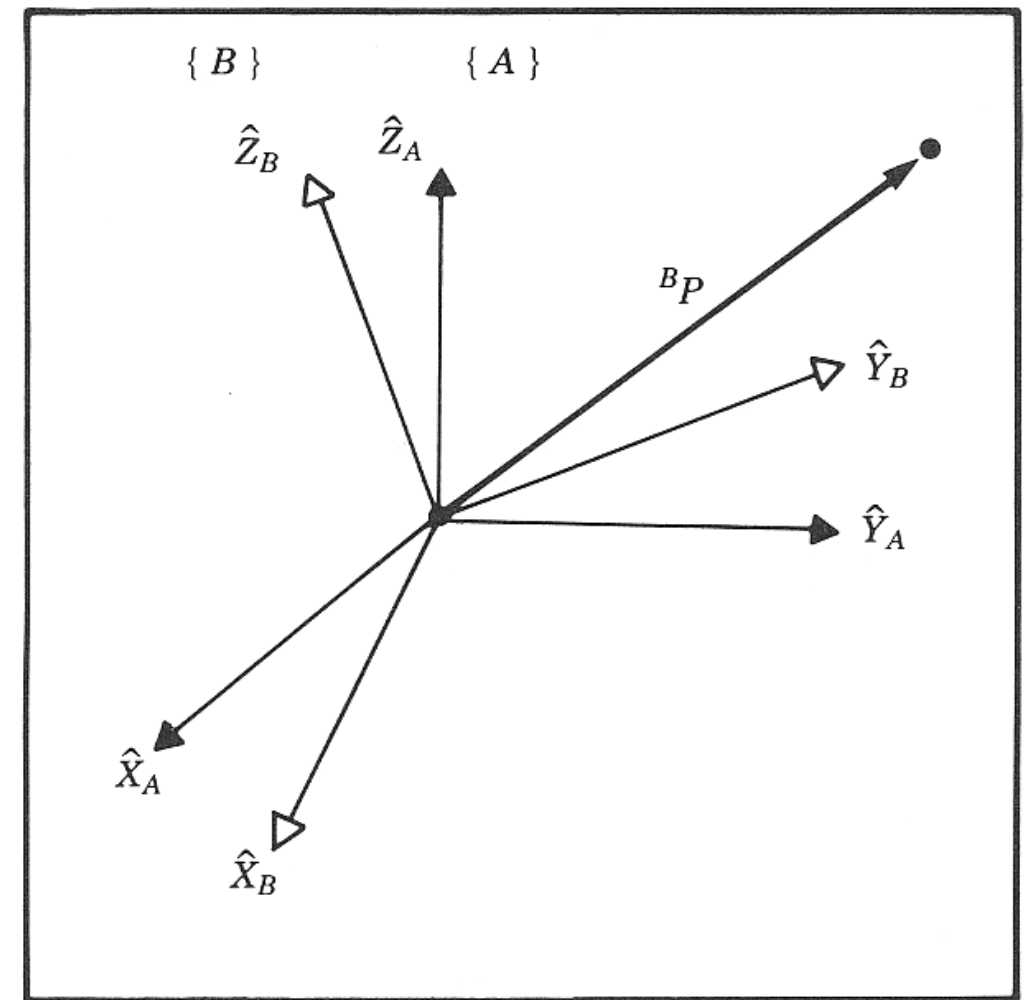
Spatial Descriptions & Transformations

Mapping: Rotated Frames

Assuming that frame {B} is *only translated* (not rotated) with respect frame {A}.
The position of the point can be expressed in frame {A} as follows.

$${}^A P = {}^A_B R \quad {}^B P$$

$${}^B P = {}^B_A R \quad {}^A P$$



Spatial Descriptions & Transformations

Mapping: Rotated Frames: Inversion

Given: The rotation matrix from frame {B} with respect to frame {A} - ${}^A_B R$

Calculate: The rotation matrix from frame {A} with respect to frame {B} - ${}^B_A R$

$${}^B_A R = {}^A_B R^{-1} = {}^A_B R^T$$

$${}^A_B R = {}^B_A R^{-1} = {}^B_A R^T$$

Spatial Descriptions & Transformations

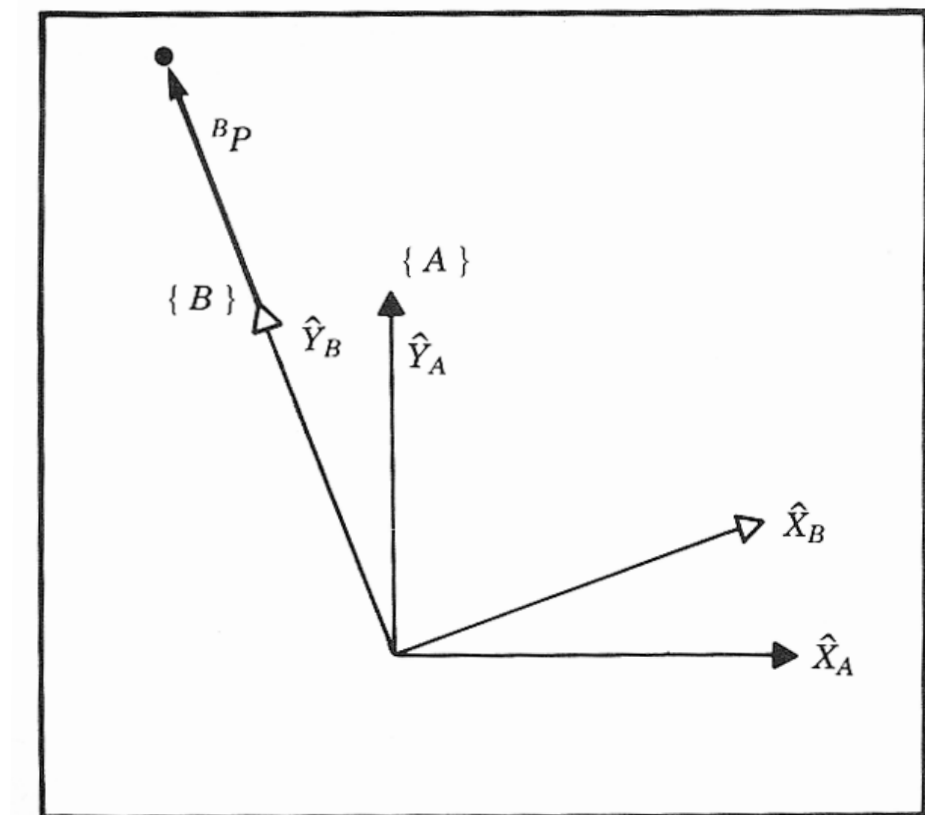
Mapping: Rotated Frames: Example

Given:

$${}^B P = \begin{bmatrix} 0 \\ {}^B p_y \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \quad \theta = 30^\circ$$

Compute: ${}^A P$

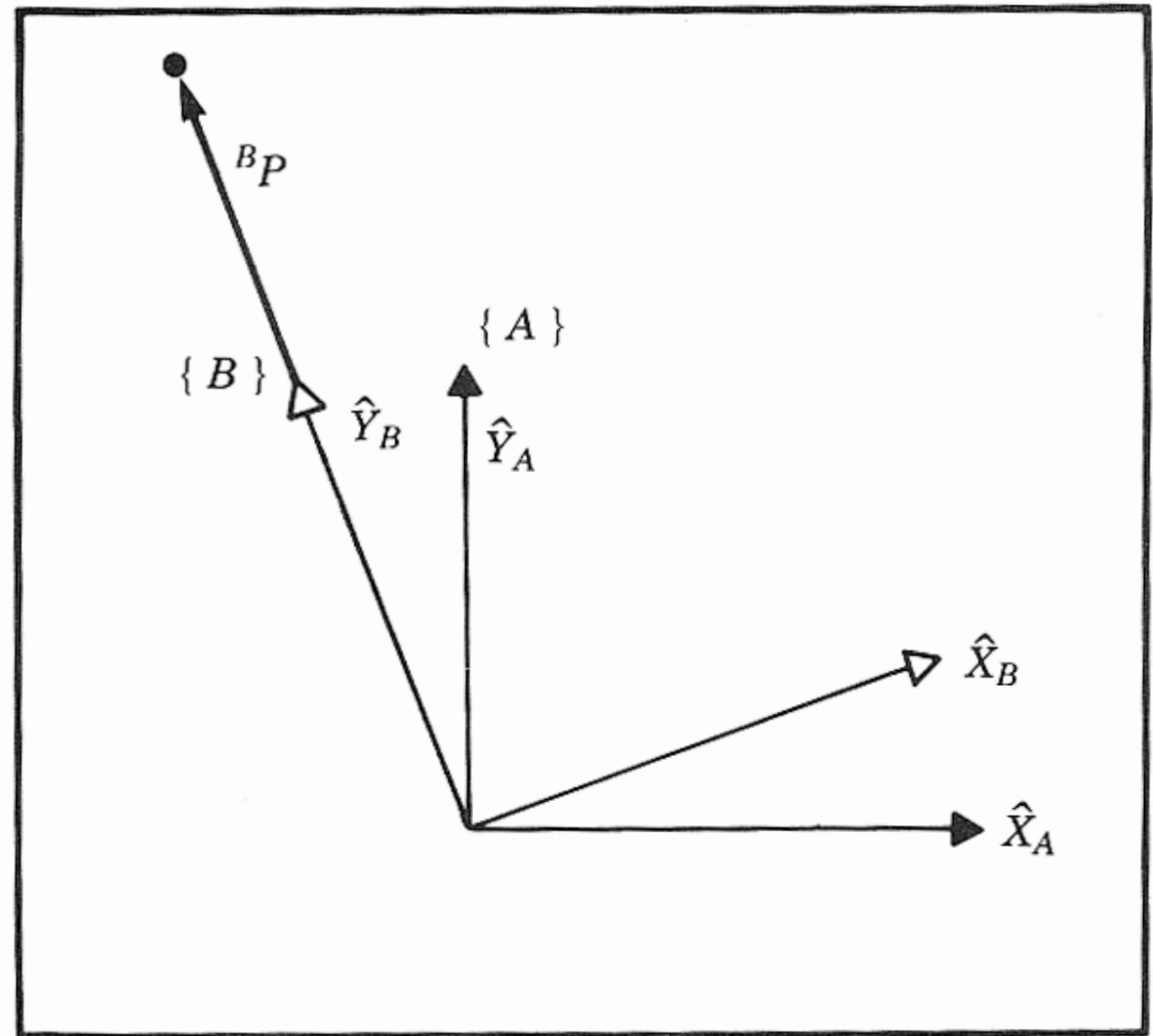
Solution: ${}^A P = {}^A R_B {}^B P$



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Mapping: Rotated Frames: Example

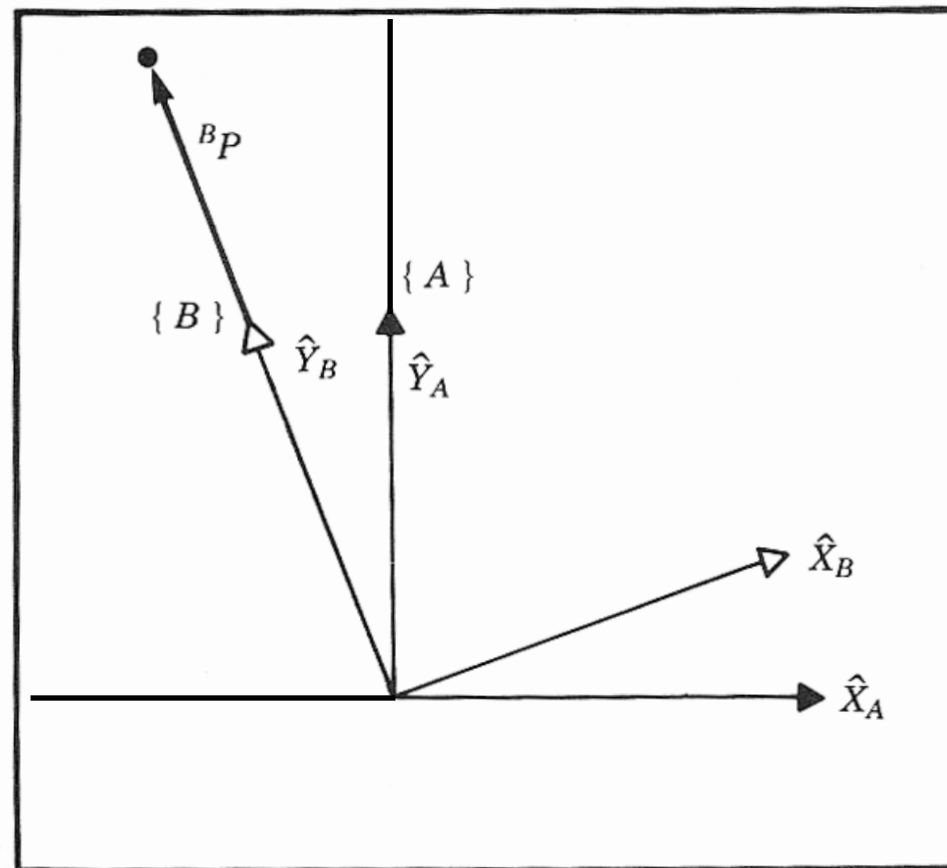
$${}^A_B R = [{}^A\hat{X}_B, {}^A\hat{Y}_B, {}^A\hat{Z}_B] =$$



Spatial Descriptions & Transformations

Mapping: Rotated Frames: Example

$${}^A P = {}^A R {}^B P = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ {}^B p_y \\ 0 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.500 & 0.000 \\ 0.500 & 0.866 & 0.000 \\ 0.000 & 0.000 & 1.000 \end{bmatrix} \begin{bmatrix} 0.000 \\ 2.000 \\ 0.000 \end{bmatrix} = \begin{bmatrix} -1.000 \\ 1.732 \\ 0.000 \end{bmatrix}$$



Spatial Descriptions & Transformations

Mapping: Rotated Frames: General Notation

The rotation matrices with respect to the reference frame are defined as follows:

$$R_X(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix}$$

$$R_Y(\beta) = \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix}$$

$$R_Z(\alpha) = \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

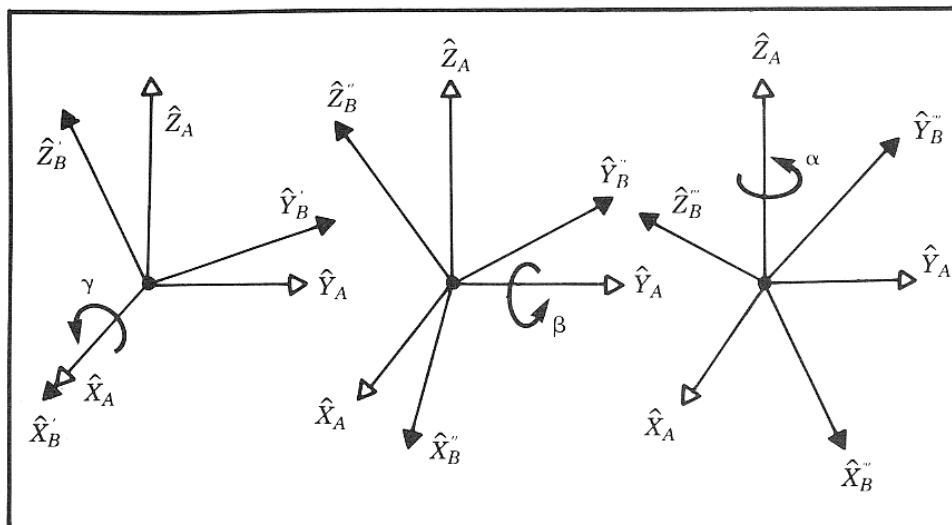
Spatial Descriptions & Transformations

Representation of Orientation: Fixed Angles, Euler Angles

- Really, we only need 3 numbers... (vs. 9 numbers in the 3X3 matrix)
- Which axis to use for the rotation? (FIXED or MOVED?)
- Order is important! (The order of the 3 axes to rotate about)

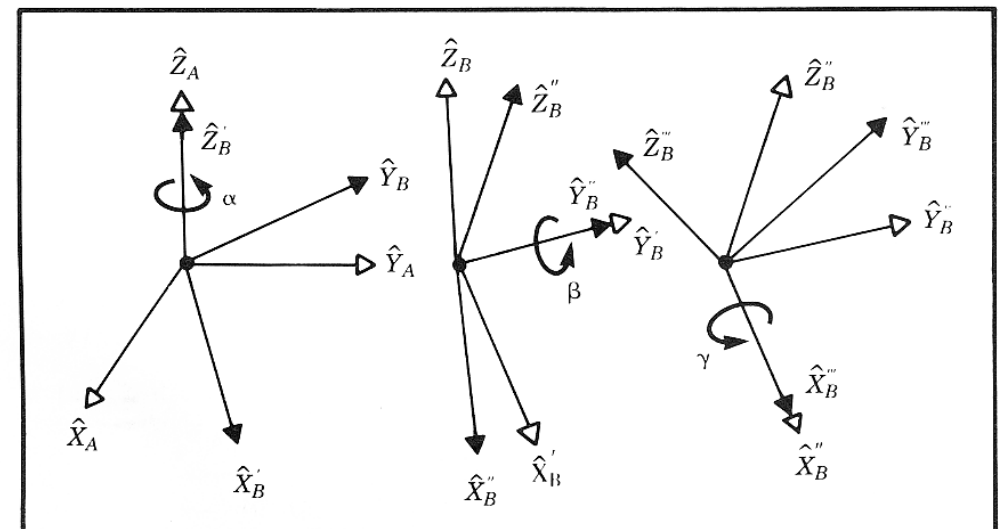
X-Y-Z Fixed Angles

The rotations perform about an axis of a **fixed** reference frame



Z-Y-X Euler Angles

The rotations are performed about an axis of a **moved** reference frame

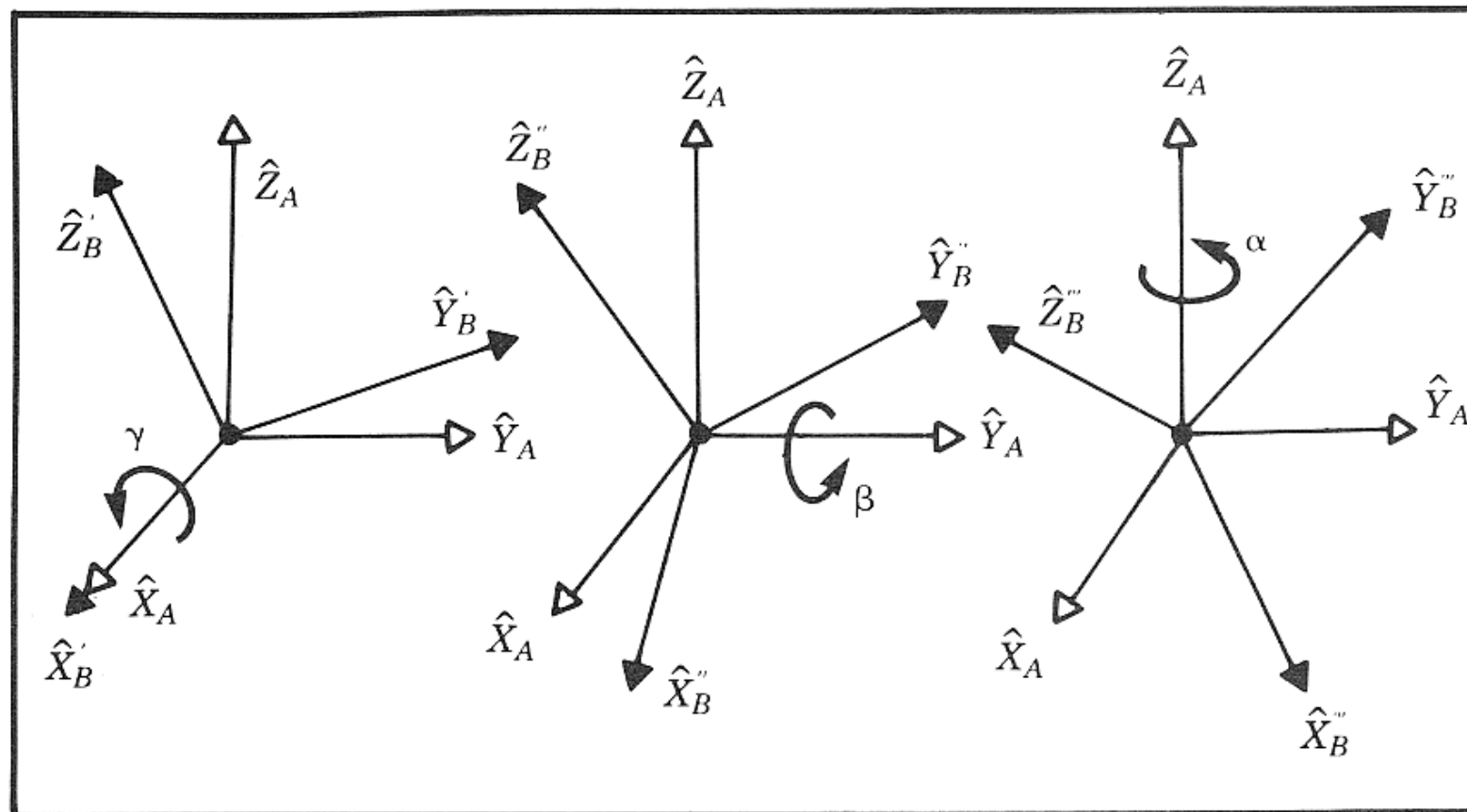


Spatial Descriptions & Transformations

Representation of Orientation: **X-Y-Z Fixed Angles**

Start with frame {B} coincident with a known reference frame {A}

- Rotate frame {B} about $\hat{X}_A$ by an angle γ
 - Rotate frame {B} about $\hat{Y}_A$ by an angle β
 - Rotate frame {B} about $\hat{Z}_A$ by an angle α
- Each rotations are performed about an axis of the **fixed** reference frame {A}



Spatial Descriptions & Transformations

Representation of Orientation: X-Y-Z Fixed Angles

When successive rotations are described with respect to *a single fixed (global) frame*, we may combine these rotations by *pre-multiplication*

$${}^A_B R_{XYZ}(\gamma, \beta, \alpha) = R_Z(\alpha) R_Y(\beta) R_X(\gamma) = \begin{matrix} \text{Z} \\ \text{3} \end{matrix} \begin{matrix} \text{Y} \\ \text{2} \end{matrix} \begin{matrix} \text{X} \\ \text{1} \end{matrix} \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix}$$

$${}^A_B R_{XYZ}(\gamma, \beta, \alpha) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$$

Spatial Descriptions & Transformations

Computing Angles from a Rotation Matrix

Given a 3x3 rotation matrix, how do we calculate the X-Y-Z fixed angles?

Given:

$${}^A_B R_{XYZ}(\gamma, \beta, \alpha) = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$$

Compute: $\gamma \quad \beta \quad \alpha$

Spatial Descriptions & Transformations

Note: $\text{Atan2}(y, x)$

Four-quadrant inverse tangent (arctangent) in the range of $[-\pi \quad \pi]$

$$\text{Atan2}(y, x) = \tan^{-1}(y / x)$$

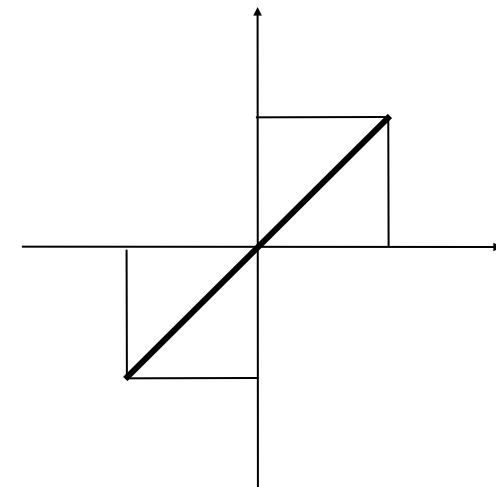
For example:

$$\text{Atan}(+1, +1) = 45^\circ$$

$$\text{Atan2}(+1, +1) = 45^\circ$$

$$\text{Atan}(-1, -1) = 45^\circ$$

$$\text{Atan2}(-1, -1) = -135^\circ$$



Spatial Descriptions & Transformations

Computing Angles from a Rotation Matrix

Given a 3x3 rotation matrix, how do we calculate the X-Y-Z fixed angles?

Given:

$${}^A_B R_{XYZ}(\gamma, \beta, \alpha) = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$$

Spatial Descriptions & Transformations

Computing Angles from a Rotation Matrix

Spatial Descriptions & Transformations

Computing Angles from a Rotation Matrix

Spatial Descriptions & Transformations

Computing Angles from a Rotation Matrix

Spatial Descriptions & Transformations

Computing Angles from a Rotation Matrix

For $-90^\circ \leq \beta \leq 90^\circ$

$$\beta = \text{Atan2} \left(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2} \right)$$

$$\alpha = \text{Atan2} \left(\frac{r_{21}}{\cos \beta}, \frac{r_{11}}{\cos \beta} \right)$$

$$\gamma = \text{Atan2} \left(\frac{r_{32}}{\cos \beta}, \frac{r_{33}}{\cos \beta} \right)$$

When $\beta = \pm 90^\circ$

We choose $\alpha = 0$

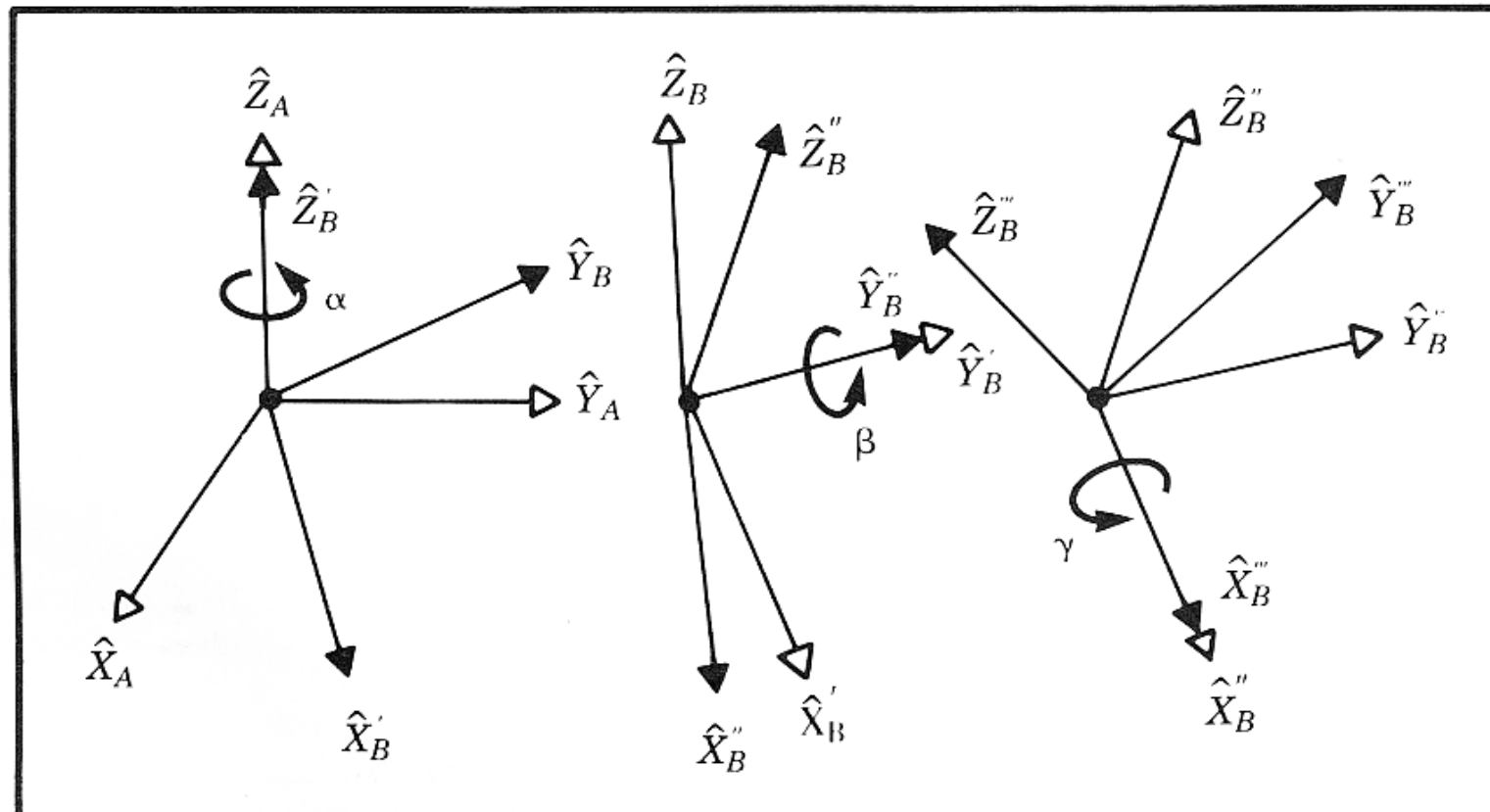
Then $\gamma = \pm \text{Atan2}(r_{12}, r_{22})$

Spatial Descriptions & Transformations

Representation of Orientation: **Z-Y-X Euler Angles**

Start with frame {B} coincident with a known reference frame {A}

- Rotate frame {B} about $\hat{Z}_A$ by an angle α
 - Rotate frame {B} about $\hat{Y}_B$ by an angle β
 - Rotate frame {B} about $\hat{X}_B$ by an angle γ
- Each rotations are performed about an axis of a **moved** reference frame



Spatial Descriptions & Transformations

Representation of Orientation: **Z-Y-X Euler Angles**

We may combine successive rotations by ***post-multiplication*** if each rotation is about a vector represented in ***the current rotated frame***

$${}^A_B R_{Z'Y'X'}(\alpha, \beta, \gamma) = R_Z(\alpha) R_Y(\beta) R_X(\gamma) = \begin{matrix} \text{Z} & \text{Y} & \text{X} \\ \text{1} & \text{2} & \text{3} \end{matrix} \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix}$$

$${}^A_B R_{Z'Y'X'}(\alpha, \beta, \gamma) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$$

Spatial Descriptions & Transformations

X-Y-Z Fixed Angles vs. Z-Y-X Euler Angles

X-Y-Z Fixed Angles

The rotations perform about an axis of a **fixed** reference frame

$${}^A_B R_{XYZ}(\gamma, \beta, \alpha) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$$

Z-Y-X Euler Angles

The rotations are perform about an axis of a **moved** reference frame

$${}^A_B R_{Z'Y'X'}(\alpha, \beta, \gamma) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$$

Hold up,
wait a minute! 🤔

Spatial Descriptions & Transformations

X-Y-Z Fixed Angles vs. Z-Y-X Euler Angles

$${}^A_B R_{XYZ}(\gamma, \beta, \alpha) = {}^A_B R_{Z'Y'X'}(\alpha, \beta, \gamma)$$

X-Y-Z Fixed Angles Z-Y-X Euler Angles

Three rotations taken about fixed axes (Fixed Angles - XYZ) yield the same final orientation as the same three rotation taken in an opposite order about the axes of the moving frame (Euler Angles ZYX)

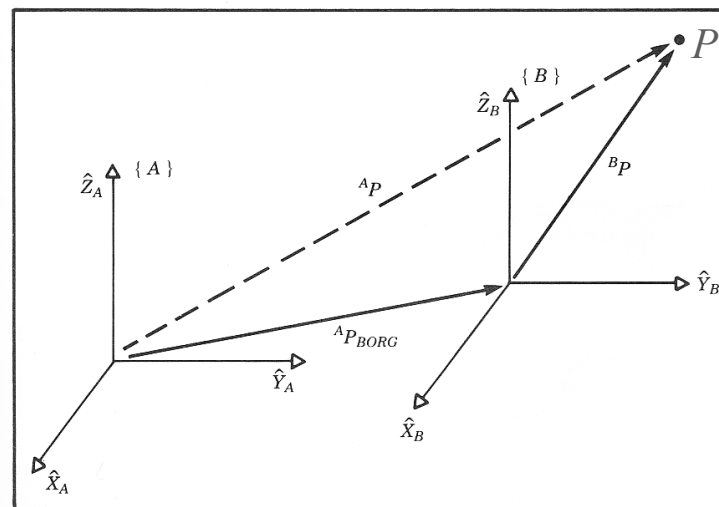


NOTE: there are 24 Angle-Set Conventions! (12 Euler angle sets and 12 fixed angle sets)
See Text Appendix B

Spatial Descriptions & Transformations

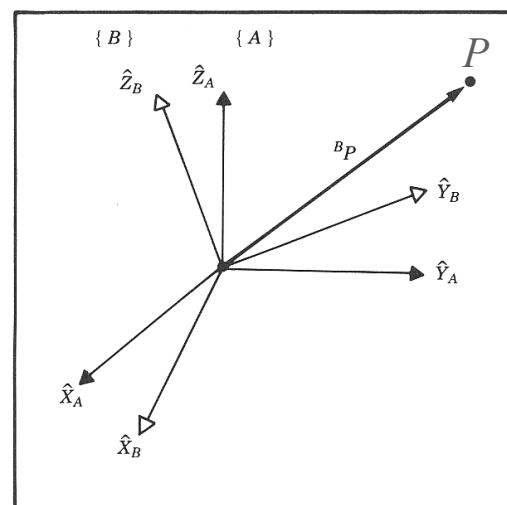
Mapping: General Frames (Translated + Rotated)

Mapping: Translated Frame



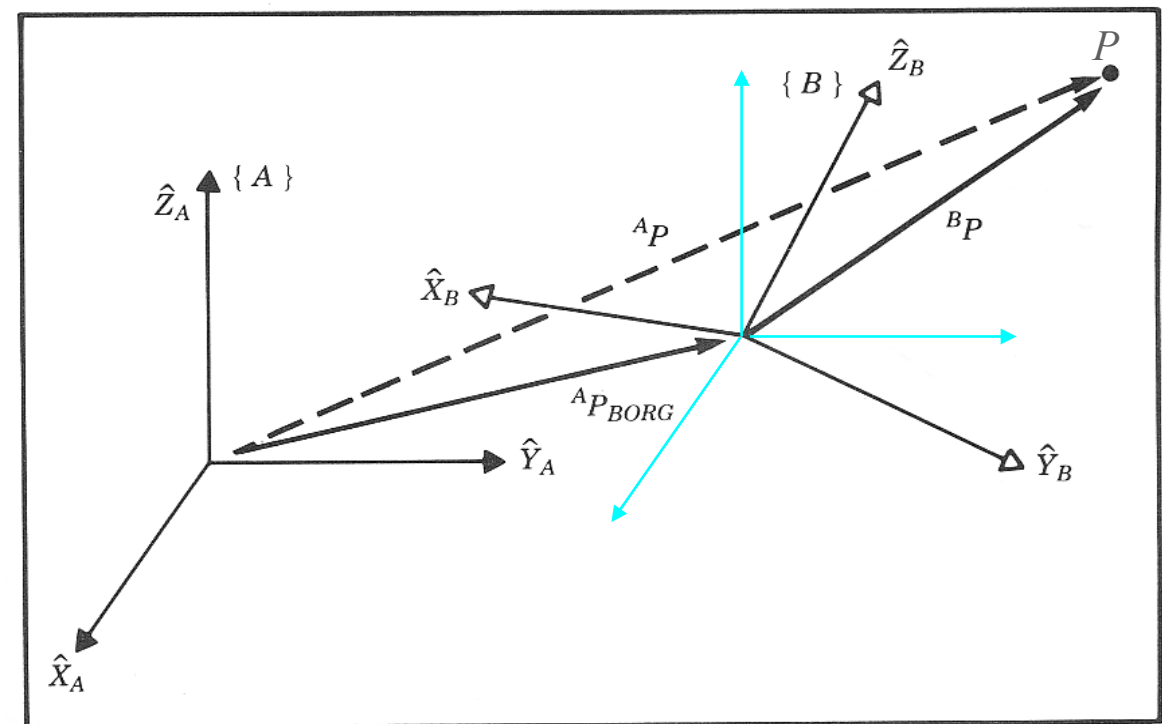
$${}^A P = {}^B P + {}^A P_{BORG}$$

Mapping: Rotated Frame



$${}^A P = {}^A R_B {}^B P$$

Mapping: Translated + Rotated Frame



$${}^A P = {}^A R_B {}^B P + {}^A P_{BORG}$$

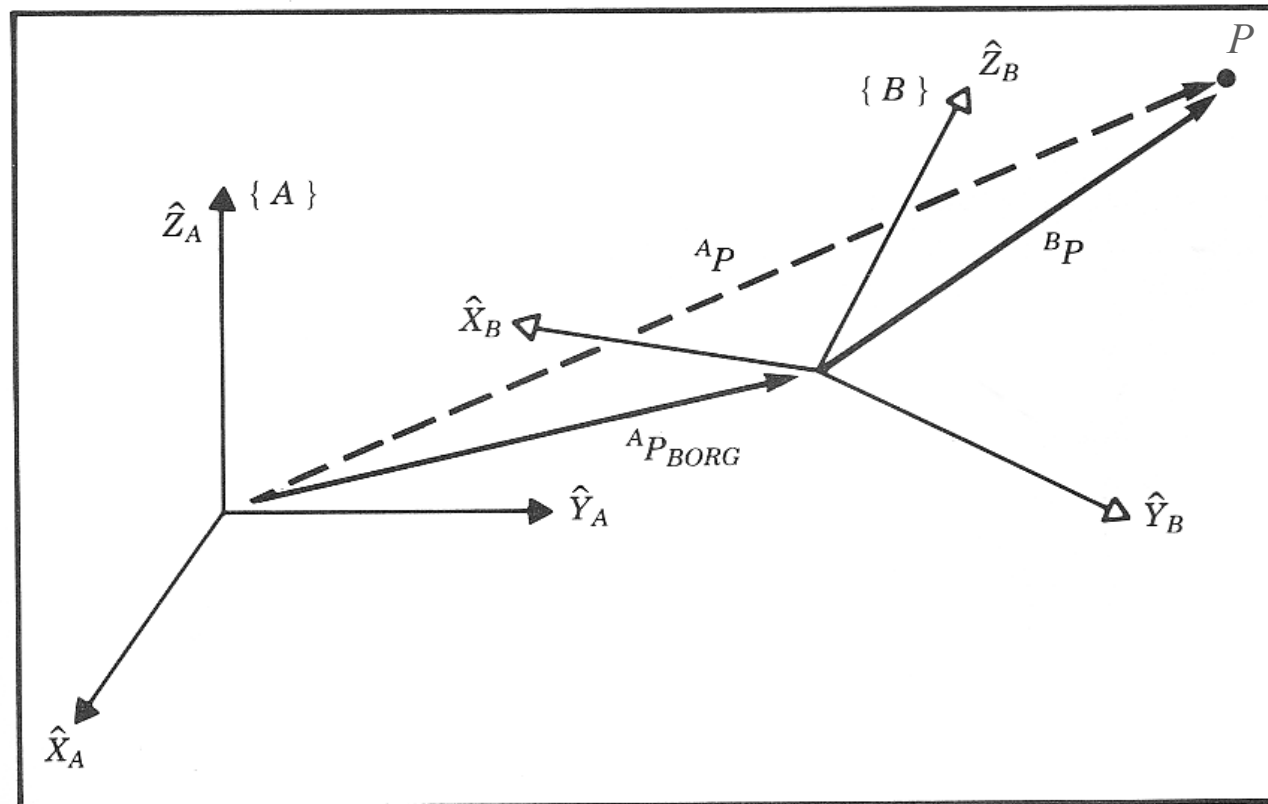
Spatial Descriptions & Transformations

Mapping: General Frames (Translated + Rotated)

Assuming that frame {B} is both translated and rotated with respect frame {A} the position of the point expressed in frame {B} can be expressed in frame {A} as follows.

$$\{B\} = \{ {}^A_B R, {}^A P_{BORG} \}$$

$${}^A P = {}^A_B R {}^B P + {}^A P_{BORG}$$



$${}^A P = {}^A_B T {}^B P \quad ? \quad \longrightarrow \quad \textit{Homogeneous Transformation!}$$

Spatial Descriptions & Transformations

Homogeneous Transformation

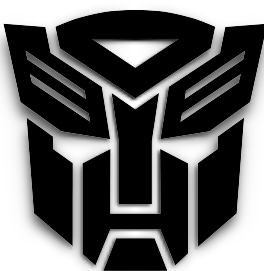
The ***homogeneous transformation matrix*** is a 4x4 matrix casting the *rotation* and *translation* of a general transform into a single matrix.

(in other fields of study it can be used to compute perspective and scaling operations.)

$${}^A P = {}^A R_B P + {}^A P_{BORG}$$

$${}^A P = {}^A T_B P$$

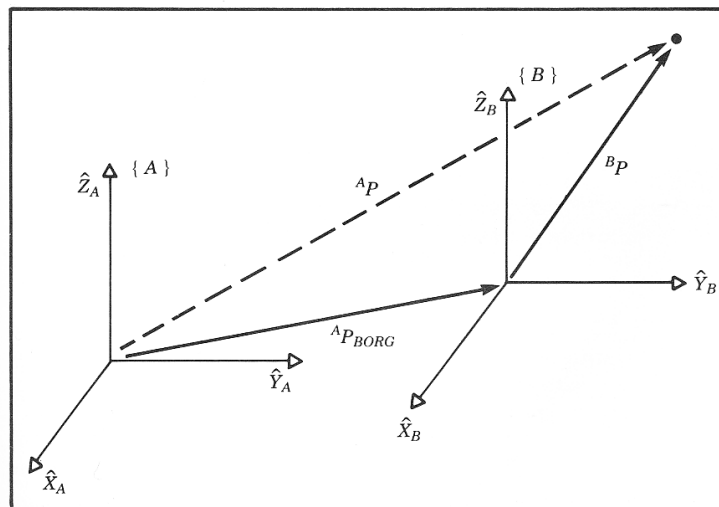
$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R_B & {}^A P_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix}$$



Spatial Descriptions & Transformations

Homogeneous Transformation: Special Cases

Translation only

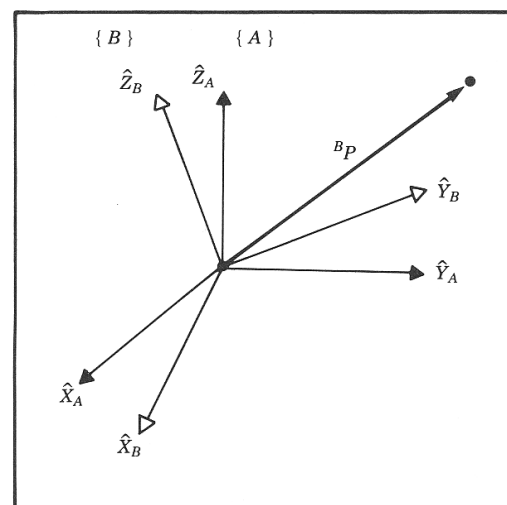


$${}^A P = {}^B P + {}^A P_{BORG}$$

$${}^A T_B = \begin{bmatrix} 1 & 0 & 0 & {}^A P_{BORGx} \\ 0 & 1 & 0 & {}^A P_{BORGy} \\ 0 & 0 & 1 & {}^A P_{BORGz} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^A P = {}^A T_B {}^B P$$

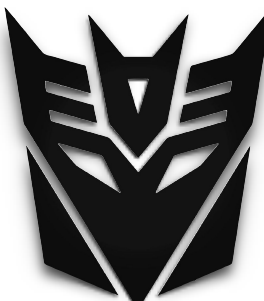
Rotation Only



$${}^A P = {}^A R_B {}^B P$$

$${}^A T_B = \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^A P = {}^A T_B {}^B P$$



Spatial Descriptions & Transformations

Homogeneous Transformation: Example

Given ${}^B P = \begin{bmatrix} {}^B p_x \\ {}^B p_y \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \\ 0 \end{bmatrix}$

Frame {B} is rotated relative to frame {A} about $\hat{Z}$ by 30 degrees, and translated
10 units in $\hat{X}_A$ and 5 units in $\hat{Y}_A$

Calculate: The vector ${}^A P$ expressed in frame {A}.

Spatial Descriptions & Transformations

Homogeneous Transformation: Example

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Homogeneous Transformation: Example

$${}^A P = {}^A T^B P = \begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R & {}^A P_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix}$$

$${}^A P = \begin{bmatrix} 0.866 & -0.500 & 0.000 & 10.0 \\ 0.500 & 0.866 & 0.000 & 5.0 \\ 0.000 & 0.000 & 1.000 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3.0 \\ 7.0 \\ 0.0 \\ 1 \end{bmatrix} = \begin{bmatrix} 9.098 \\ 12.562 \\ 0.0 \\ 1 \end{bmatrix}$$

Spatial Descriptions & Transformations

Operator: Translation, Rotation, Transformation

The same mathematical forms used for **mapping** can also be used as **operators** to translate points, rotate vectors or do both

Translation Operator

Operates on a vector ${}^A P_1$ and changes that vector to a new vector ${}^A P_2$, by means of a translation

$${}^A P_2 = {}^A P_1 + {}^A Q$$

Rotation Operator

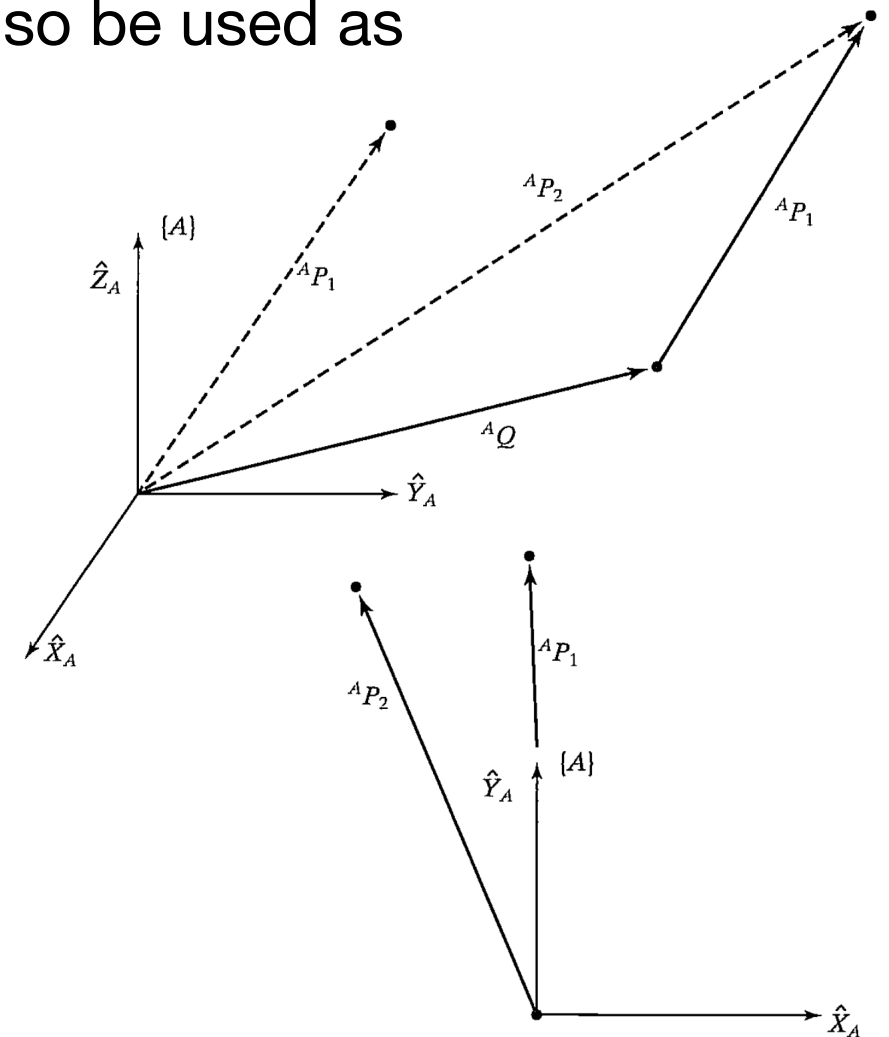
Operates on a vector ${}^A P_1$ and changes that vector to a new vector ${}^A P_2$, by means of a rotation

$${}^A P_2 = R {}^A P_1$$

Transformation Operator

Operates on a vector ${}^A P_1$ and changes that vector to a new vector ${}^A P_2$, by means of both rotation and translation

$${}^A P_2 = T {}^A P_1$$



Spatial Descriptions & Transformations

Operator: Rotation Example

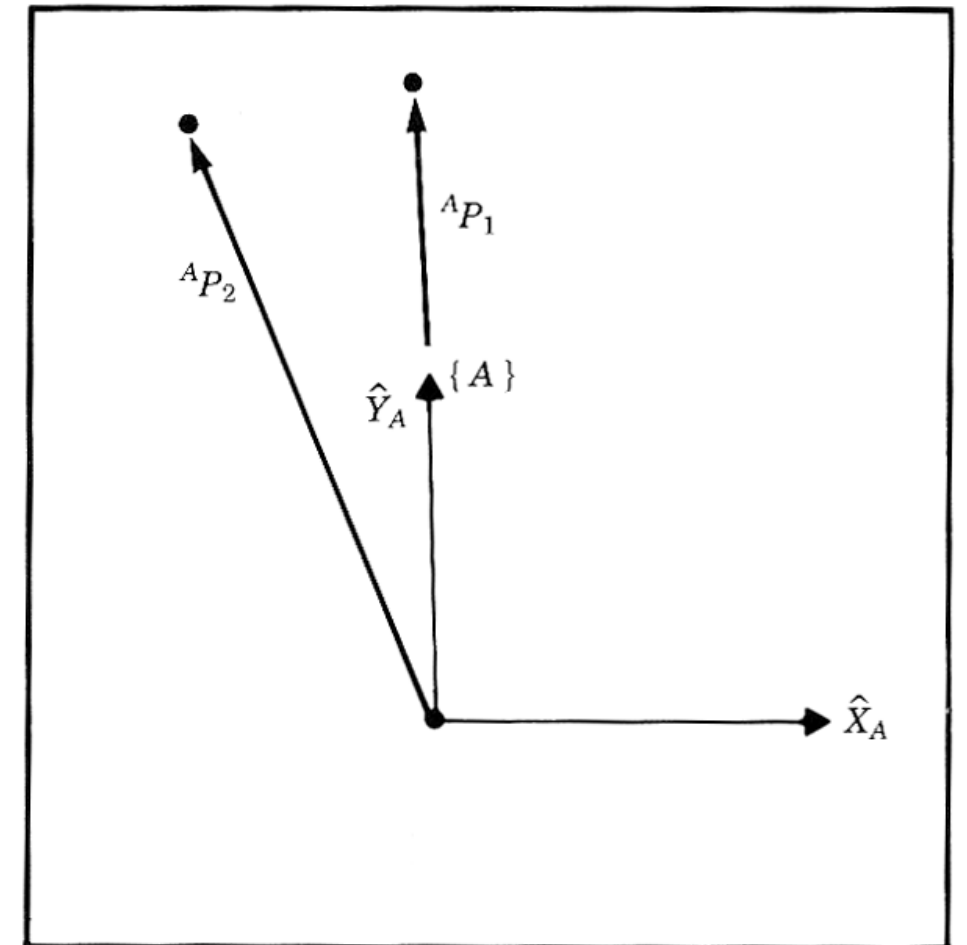
Given: ${}^A P_1 = \begin{bmatrix} 0 \\ {}^A p_{1y} \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$

Compute:

The vector ${}^A P_2$ obtained by rotating this vector about $\hat{Z}$ by 30 degrees

Solution:

$${}^A P_2 = R(30^\circ) {}^A P_1 = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ {}^A p_{1y} \\ 0 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.500 & 0.000 \\ 0.500 & 0.866 & 0.000 \\ 0.000 & 0.000 & 1.000 \end{bmatrix} \begin{bmatrix} 0.000 \\ 2.000 \\ 0.000 \end{bmatrix} = \begin{bmatrix} -1.000 \\ 1.732 \\ 0.000 \end{bmatrix}$$



Spatial Descriptions & Transformations

Homogeneous Transformation: Interpretation

As a general tool to represent a frame we have introduced the homogeneous transformation, a 4x4 matrix containing orientation and position information.

Three interpretation of the homogeneous transformation:

1. **Description of a Frame:** ${}^A_B T$ describes frame {B} relative to frame {A} $\{B\}$
2. **Transform Mapping:** ${}^A_B T$ maps P from frame {B} wrt frame {A} ${}^B P \rightarrow {}^A P$
3. **Transform Operator:** T operates on ${}^A P_1$ to create ${}^A P_2$ ${}^A P_2 = T {}^A P_1$

$${}^A_B T = \begin{bmatrix} {}^A_B R & {}^A P_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Spatial Descriptions & Transformations

Compound Transformations

Given: Vector ${}^C P$
 Frame {C} is known relative to frame {B} ${}^B_C T$
 Frame {B} is known relative to frame {A} ${}^A_B T$

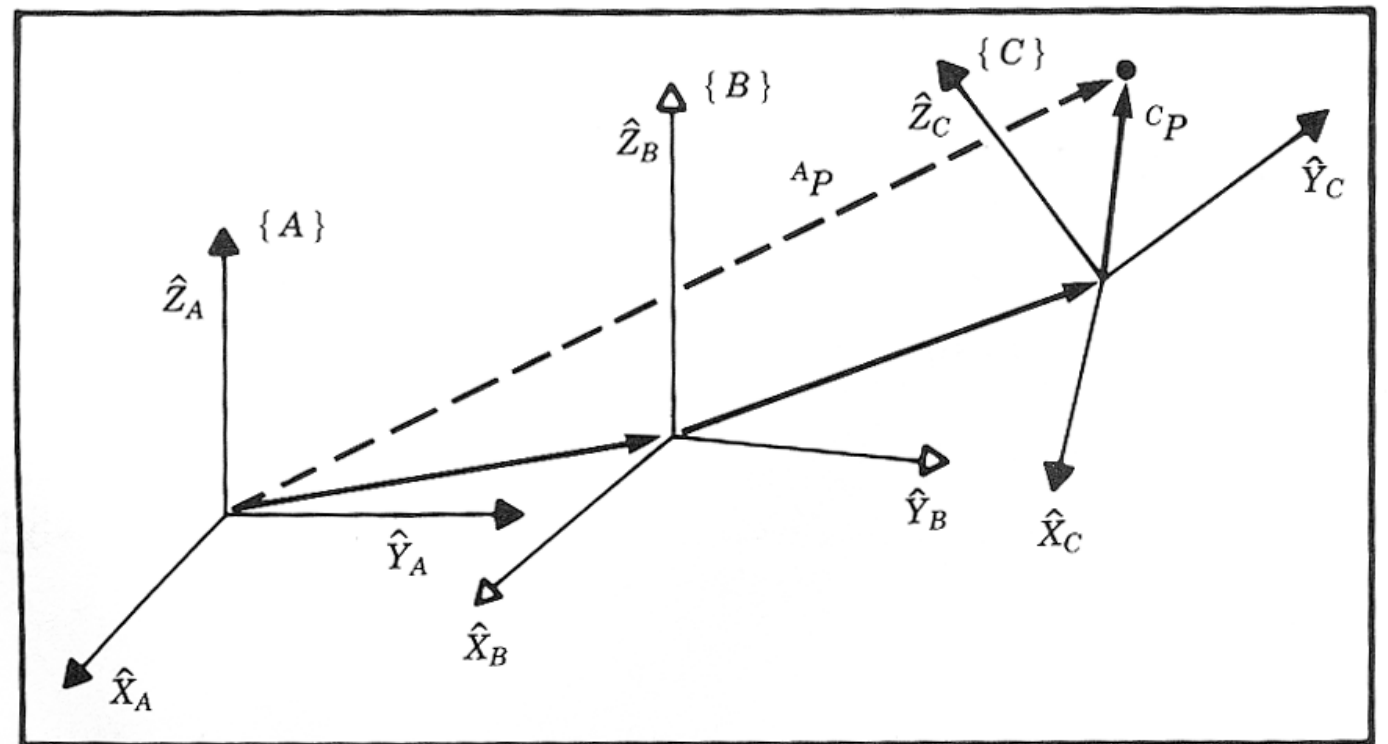
Calculate: Vector ${}^A P$

Solution:

$${}^B P = {}^B_C T {}^C P$$

$${}^A P = {}^A_B T {}^B P$$

$${}^A P = {}^A_B T {}^B_C T {}^C P$$



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Inverting a Transform

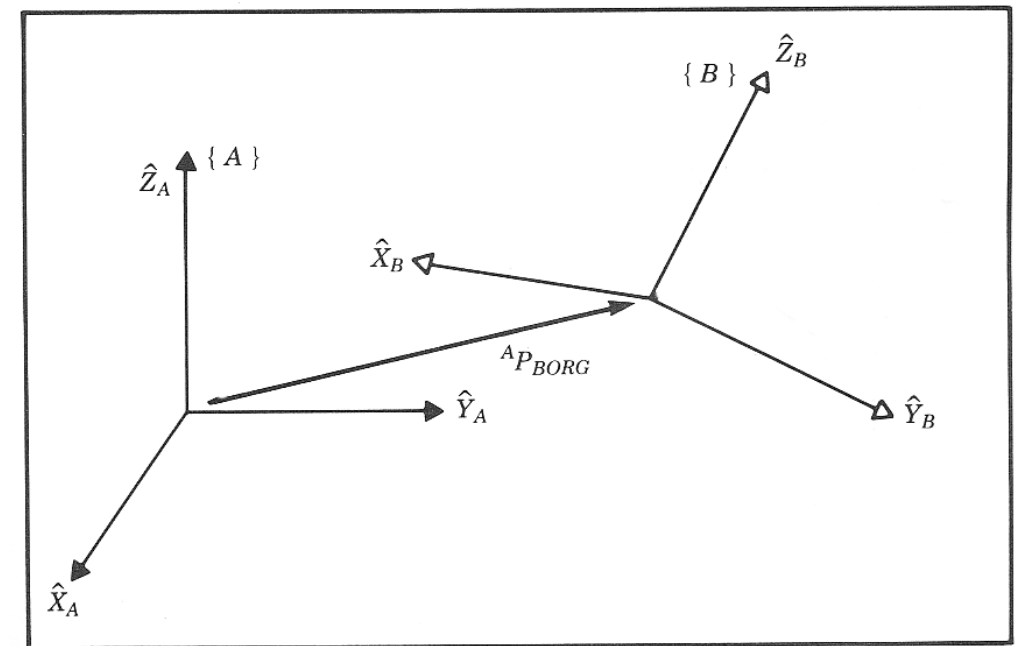
Given: Description of frame {B} relative to frame {A} ${}^A T_B \quad ({}^A R, {}^A P_{BORG})$

Calculate: Description of frame {A} relative to frame {B} ${}^B T_A \quad ({}^B R, {}^B P_{AORG})$

$${}^B R_A = {}^A R_B^T$$

$${}^B P_{AORG} = -{}^B R_A {}^A P_{BORG} = -{}^A R_B^T {}^A P_{BORG}$$

$${}^B T_A = \begin{bmatrix} {}^A R_B^T & -{}^A R_B^T {}^A P_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Note: ${}^B T_A = {}^A T_B^{-1}$

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Inverting a Transform: Example

Given: Description of frame {B} relative to frame {A} ${}^A T_B ({}^A R, {}^A P_{BORG})$
Frame {B} is rotated relative to frame {A} about $\hat{Z}$ by 30 degrees,
and translated 4 units in $\hat{X}$, and 3 units in $\hat{Y}$

Calculate: Homogeneous Transform ${}^B T_A ({}^B R, {}^B P_{AORG})$

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Inverting a Transform: Example

Solution:

$${}^A_B T = \begin{bmatrix} c\theta & -s\theta & 0 & {}^A P_{BORGx} \\ s\theta & c\theta & 0 & {}^A P_{BORGy} \\ 0 & 0 & 1 & {}^A P_{BORGz} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.500 & 0.000 & 4.000 \\ 0.500 & 0.866 & 0.000 & 3.000 \\ 0.000 & 0.000 & 1.000 & 0.000 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

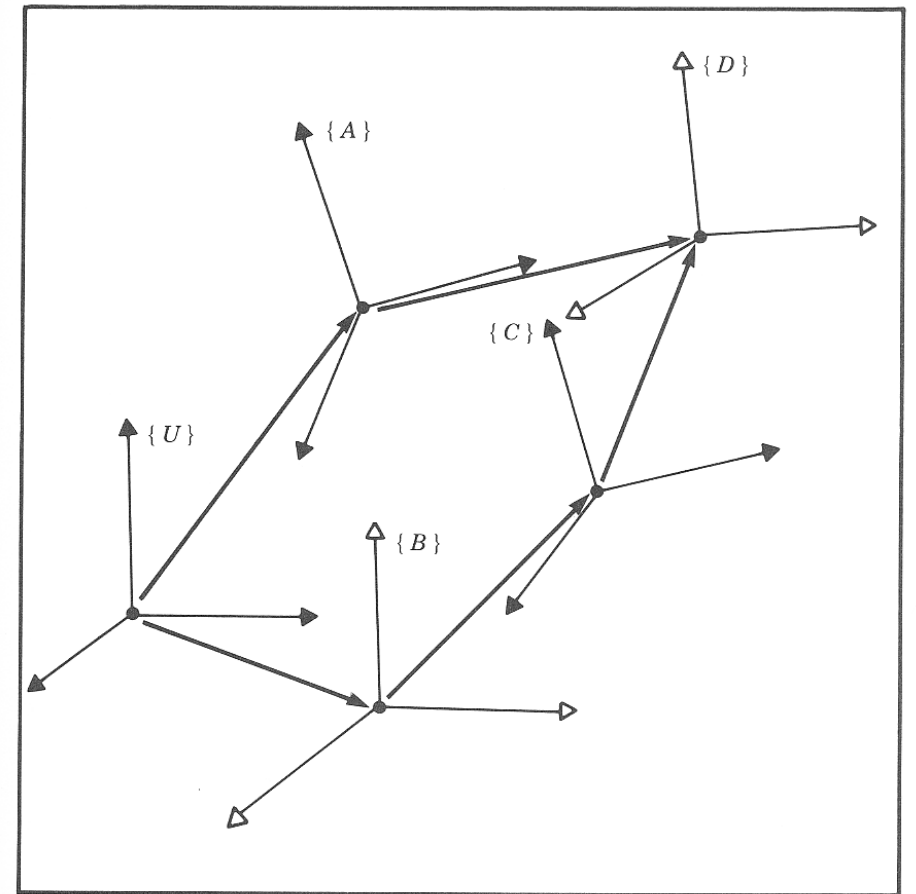
$${}^B_A T = \begin{bmatrix} {}^A_B R^T & -{}^A_B R^T {}^A P_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.866 & 0.500 & 0.000 & -4.964 \\ -0.500 & 0.866 & 0.000 & -0.598 \\ 0.000 & 0.000 & 1.000 & 0.000 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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Transform Equations

Given: ${}^U_A T$, ${}^A_D T$, ${}^U_B T$, ${}^C_D T$

Calculate: ${}^B_C T$



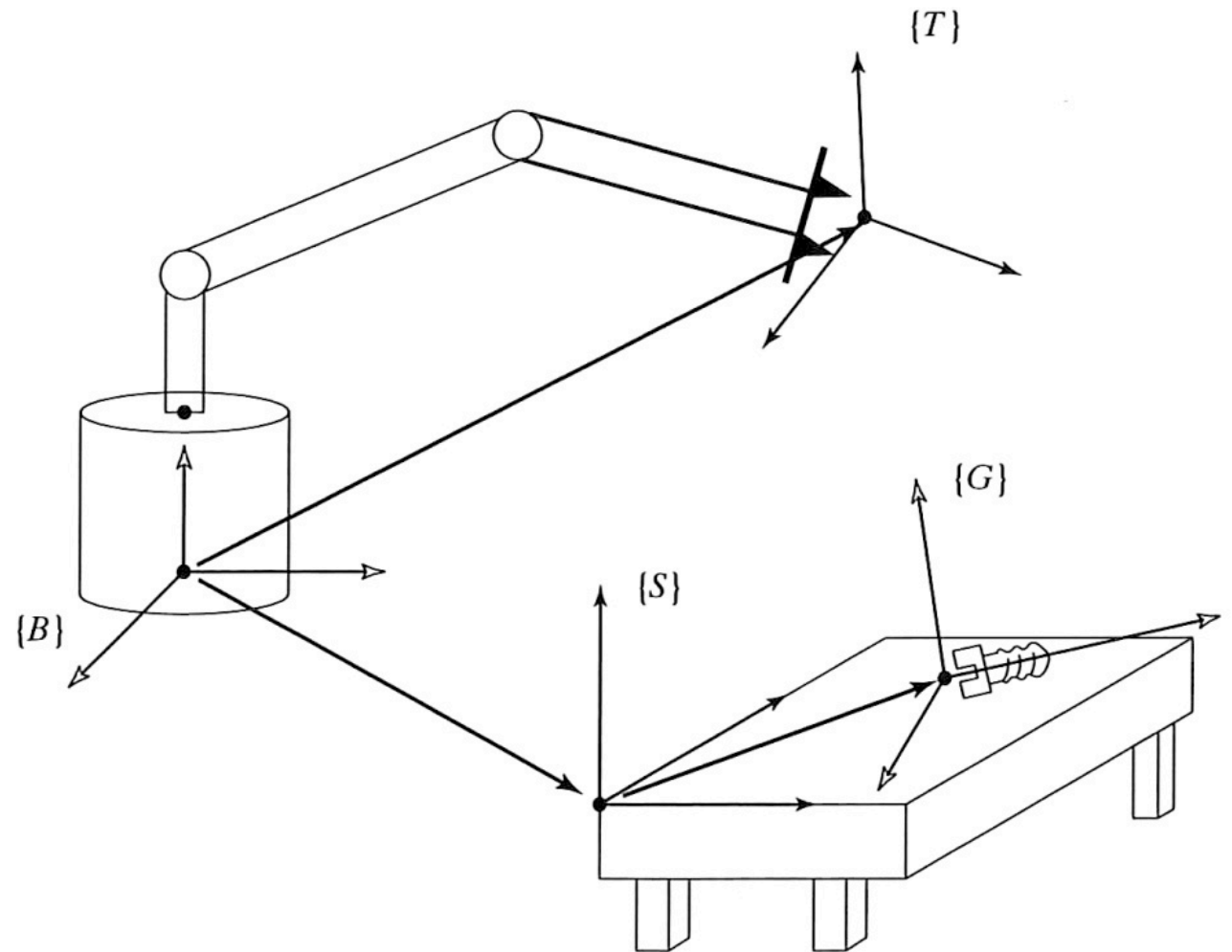
Spatial Descriptions & Transformations

Transform Equations

Given: ${}^B_T T$, ${}^S_T T$, ${}^S_G T$

Calculate: ${}^T_G T$

Solution:



$${}^T_G T = {}^B_T T^{-1} {}^B_S T {}^S_G T$$